

11. Compound **X** is thought to be a halogenoalkane containing only one halogen atom.

(a) Describe a chemical test by which you could identify which halogen is present in **X**. [4]

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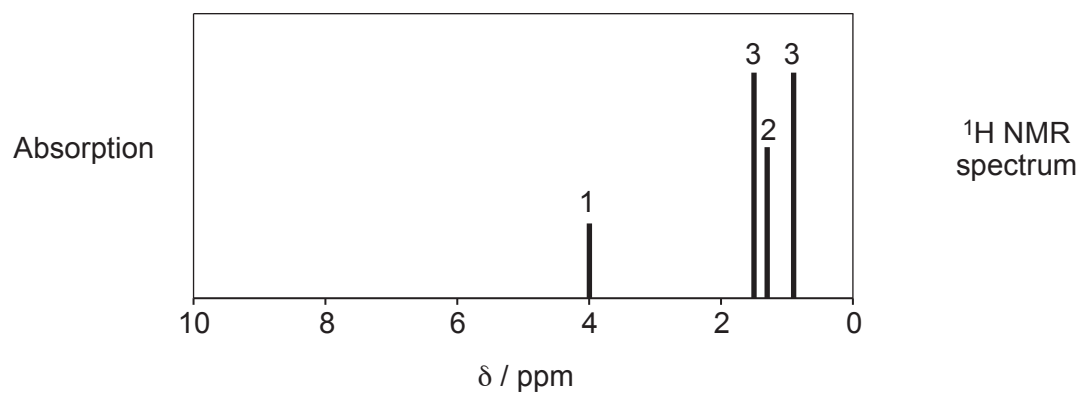
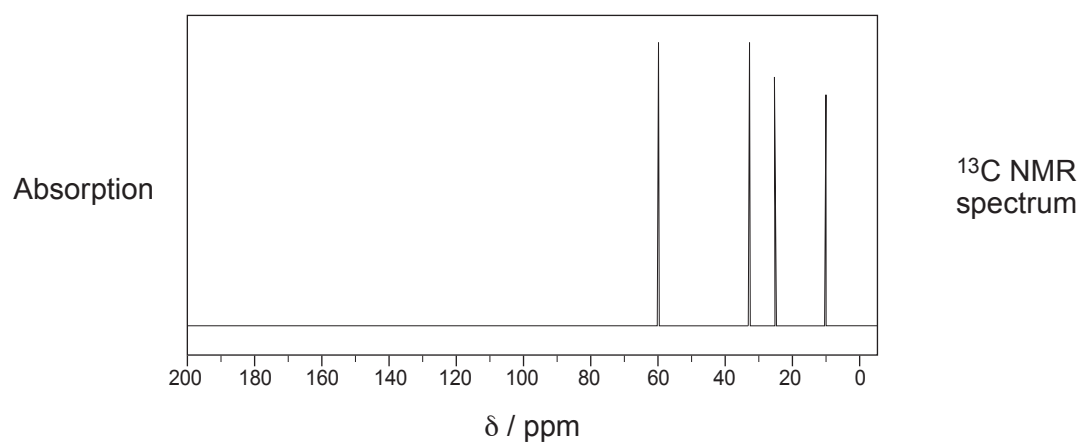
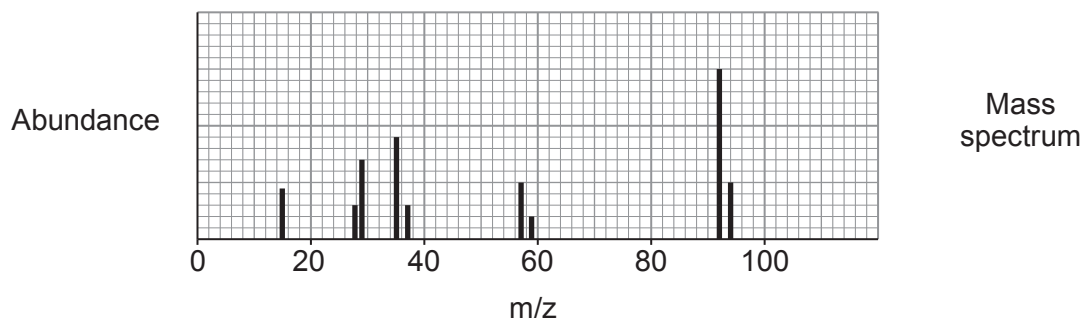
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- (b) A simplified form of the mass spectrum, the  $^{13}\text{C}$  NMR spectrum and the low resolution  $^1\text{H}$  NMR spectrum of halogenoalkane **X** are shown.



Use this information to find the structure of halogenoalkane **X**. Show your reasoning.

You must use information from **all** the spectra.

[8]

Examiner  
only

### Structure of **X**

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